Game Theory in Electricity Markets

Week 2 – Solution Concepts 1

Thank you, Marcel!



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Strategic games in normal form

Definition. A *strategic game in normal form* is a tuple

$$\Gamma = (N, (A_i)_{i \in N}, (c_i)_{i \in N})$$

where:

N finite, non-empty set of players

$N = \{1, \ldots, n\}$: the set of players participating in the game.

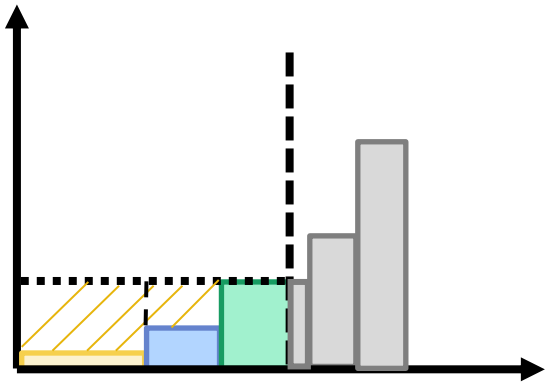
A_i non-empty set of actions for player i

The set of actions available to player $i \in N$. We write $A = A_1 \times A_2 \times \cdots \times A_n$ for the set of joint actions (action profiles), and $a = (a_1, \ldots, a_n) \in A$ for a typical joint action.

$c_i : A \to \mathbb{R}$ cost function for player i

Maps a joint action $a \in A$ to the cost $c_i(a)$ incurred by player i.

The market clearing game



Let's instantiate $\Gamma = (N, (A_i), (c_i))$ for last week's market clearing problem:

$N = \{G_1, \ldots, G_5\}$ the five generators

Each of the five generators from last week's market-clearing example (G_1–G_5) is a player.

$A_i = [0, \text{bid_ub}_i]$ generator i's bid

Generator i's action is the price it bids for its capacity.

$c_i(a) = c_i \cdot \text{dispatch}_i(a) - \text{price}(a) \cdot \text{dispatch}_i(a)$ generator i's cost under bids a

True production cost minus market revenue: the area between generator i's true-cost block and the market-clearing price, over the quantity it is dispatched — the same “profit” area from last week, now written as a cost.

Dominant and dominated strategies

Dominant strategy

A strategy $a_i^* \in A_i$ is a dominant strategy for player i if it gives i the lowest possible cost no matter what the other players do:

$$c_i(a_i^*, a_{-i}) \leq c_i(a_i, a_{-i})$$

for every $a_i \in A_i$ and every $a_{-i} \in A_{-i}$.

A dominant strategy is a best response to everything the other players might do — no guessing required.

Dominated strategies

Strictly dominated

$a_i \in A_i$ is strictly dominated if some $a_i' \in A_i$ is always strictly better:

$$c_i(a_i', a_{-i}) < c_i(a_i, a_{-i})$$

for every $a_{-i} \in A_{-i}$.

Weakly dominated

$a_i \in A_i$ is weakly dominated if some $a_i' \in A_i$ is never worse, and sometimes strictly better:

$$c_i(a_i', a_{-i}) \leq c_i(a_i, a_{-i})$$

for every $a_{-i} \in A_{-i}$, with strict inequality for at least one a_{-i}.

Checking for dominated actions

Q1 What are the dominated actions in the bed problem? Are they weakly or strictly dominated?

A1 To get the bed at any day except the first day is strictly dominated by getting it on the first day.

Q2 Are there any dominated actions in the war of attrition?

A2 No, how good an action performs always depends on what the other person does.

Q3 Are there any dominated actions in our market model?

A3 To bid below your actual cost is weakly dominated by bidding exactly your own cost.

Dominant strategy equilibrium

Dominant strategy equilibrium

An action profile $a^* = (a_1^*, \ldots, a_n^*) \in A$ is a dominant strategy equilibrium if a_i^* is a dominant strategy for every player $i \in N$:

$$c_i(a_i^*, a_{-i}) \leq c_i(a_i, a_{-i})$$

for every $i \in N$, every $a_i \in A_i$, and every $a_{-i} \in A_{-i}$.

No player can do better by deviating, no matter what anyone else does.

The Prisoner's Dilemma

	P2: Silent	P2: Confess
P1: Silent	2, 2	10, 0
P1: Confess	0, 10	9, 9

Confess strictly dominates Silent for both prisoners, so (Confess, Confess) is the dominant-strategy equilibrium.

Elimination of strictly dominated actions

- Rationality tells us that no one will ever play a strictly dominated action, as they would always be better of by playing the action that strictly dominates it.
- We may therefore eliminate them from the game as they are not adding anything

The Prisoner's Dilemma 2

	P2: Silent	P2: Confess	P2: Attack
P1: Silent	2, 2	10, 0	2, 100
P1: Confess	0, 10	9, 9	10, 100
P1: Attack	100, 2	100, 10	100, 100

No dominant strategy equilibrium anymore. We can eliminate dominated actions. Afterwards we have a dominant strategy equilibrium. We say the game is dominant solvable.

Hawk and Dove

Consider Hawk and Dove with V = 10 and C = 8.
Payoff matrix:

	H	D
H	-3, -3	9, 1
D	1, 9	5, 5

Is there a dominant strategy equilibrium?

No!

There is none in Rock, Paper, Scissors

Neither in the war of attrition

Neither in our market model

→ They are very rare. □

→ We need a better solution concept!

Imagine a drumroll and Gregorian chants.

(pure) Nash equilibrium

Definition. An action profile $a^* = (a_1^*, \ldots, a_n^*) \in A$ is a *Nash equilibrium* if no player can lower their own cost by unilaterally deviating:

$$c_i(a_i^*, a_{-i}^*) \leq c_i(a_i, a_{-i}^*)$$

for every player $i \in N$ and every $a_i \in A_i$.

Weaker than a dominant strategy equilibrium: here a_i^* only depends on the other players' equilibrium actions a_{-i}^*, it does not need to be optimal for everything they might do.

Nash equilibrium in action

Consider Hawk and Dove with V = 10 and C = 8.
Payoff matrix:

	H	D
H	-3, -3	9, 1
D	1, 9	5, 5

Is there a Nash equilibrium?

Yes, even two!

Consider war of attrition with $v_1 < v_2$.

What are the Nash equilibria?

Nash equilibrium in action

Consider the war of attrition with $v_1 < v_2$. The Nash equilibria are

$$a_1 = 0, \qquad a_2 > v_1$$

No proof, just intuition:

Why someone must bid 0

Whoever is losing pays their own bid and gets nothing for it. Lowering that bid down to 0 doesn't change the outcome but strictly reduces their cost. In any equilibrium, the lower bidder must bid exactly 0.

Why a_2 must exceed v_1

With $a_i = 0$, player j wins and pays nothing at all, so any $a_j > 0$ is equally good for them. The only remaining question is player i: could they gain by overbidding a_j and winning instead? That would yield them $v_i - a_j$ which beats their current cost of 0 only if $a_j < v_i$. The equilibrium survives exactly when $a_j > v_i$. This only works when i=1 and j=2.

Best response

Definition. For player i and given the other players' actions $a_{-i} \in A_{-i}$, player i's *best-response set* is

$$BR_i(a_{-i}) = \{ a_i \in A_i : c_i(a_i, a_{-i}) \leq c_i(a_i', a_{-i}) \text{ for all } a_i' \in A_i \}$$

the action(s) in A_i that minimize player i's cost, holding everyone else's actions fixed.

Nash equilibrium, restated

$a^* \in A$ is a Nash equilibrium *if and only if*

$$a_i^* \in BR_i(a_{-i}^*) \quad \text{for every } i \in N.$$

Finding an equilibrium is the same as finding a profile where every player is simultaneously best-responding to everyone else.

Best response algorithm

- We saw that not every game has a pure Nash equilibrium
- To check if a game has a pure NE is exponential in the number of players
- Even when we know there must exist one, finding it is exponential in the number of players

⟶ Generally, we know nothing.

But there is hope:

Best-response algorithm

```
Input: any action profile a = (a_1, …, a_n)
repeat
    for each player i ∈ N:
        a_i ← BR_i(a_-i)
until no player's action changes
return a
```

If it converges, we found a Nash equilibrium.

If it does not converge, we know nothing!

Best response algorithm

Rock, Paper, Scissors

	R	P	S
R	0, 0	-1, 1	1, -1
P	1, -1	0, 0	-1, 1
S	-1, 1	1, -1	0, 0

There is no pure Nash equilibrium, so it will never converge.

Rock, Paper, Scissors, Kiss

	R	P	S	K
R	0, 0	-1, 1	1, -1	0, -10
P	1, -1	0, 0	-1, 1	0, -10
S	-1, 1	1, -1	0, 0	0, -10
K	-10, 0	-10, 0	-10, 0	**100, 100**

If we start by playing actual R,P,S the algorithm never stops playing.

A pure Nash exists. But it can't be found if the starting point is wrong.

If the best-response algorithm does not converge, we know nothing!

ε-Nash equilibrium

ε-Nash equilibrium

Definition. Fix a tolerance $\varepsilon \geq 0$. An action profile $a^* = (a_1^*, \ldots, a_n^*) \in A$ is an *ε-Nash equilibrium* if no player can lower their own cost by more than ε by unilaterally deviating:

$$c_i(a_i^*, a_{-i}^*) \leq c_i(a_i, a_{-i}^*) + \varepsilon$$

for every player $i \in N$ and every $a_i \in A_i$.

Every Nash equilibrium is a 0-NE. Allowing ε > 0 only enlarges the set of qualifying profiles. This is useful when an algorithm can only guarantee approximately-best responses.

ε-best-response algorithm

```
Input: a = (a_1, …, a_n),   ε ≥ 0
repeat
  # 1. compute all best responses
  for each player i ∈ N:
      b_i ← BR_i(a_-i)
  # 2. Check if the ε-condition fails
  for any player j ∈ N:
      if c_i(a_i, a_-i) > c_i(b_i, a_-i) + ε for
         any i:
                    a_j ← b_j
until no player's cost changes enough
return a
```

Calculate all best response first – then update

DTU

This helps finding a Nash equilibrium. It does not help if there is none.

Mixed strategies

So far, player i chose a single action $a_i \in A_i$ with certainty. A *mixed strategy* instead lets player i randomize over their actions:

$\sigma_i \in \Delta(A_i)$ player i's mixed strategy

A probability distribution over player i's action set A_i, assigning probability $\sigma_i(a_i)$ to each pure action $a_i \in A_i$.

$\sigma = (\sigma_1, \ldots, \sigma_n)$ the joint mixed strategy

We assume the players randomize independently, so σ is the product distribution: $\sigma(a) = \sigma_1(a_1) \cdot \sigma_2(a_2) \cdots \sigma_n(a_n)$, for every joint action $a = (a_1, \ldots, a_n) \in A$.

Player i's *expected cost* under σ is then defined by the integral

$$c_i(\sigma) = \int_A c_i(a) \, d\sigma(a)$$

the expectation of the (deterministic) cost $c_i(a)$ over joint actions are drawn according to the independent product distribution σ.

Mixed Nash equilibrium

Definition. A mixed profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a *mixed Nash equilibrium* if no player can lower their expected cost by unilaterally deviating:

$$c_i(\sigma_i^*, \sigma_{-i}^*) \leq c_i(\sigma_i, \sigma_{-i}^*)$$

for every player $i \in N$ and every $\sigma_i \in \Delta(A_i)$.

Recall player i's expected cost, now written as an *expected value* rather than an integral:

$$c_i(\sigma) = \sum_{a \in A} \sigma(a)\, c_i(a) = E_{a \sim \sigma}\,[c_i(a)]$$

A pure Nash equilibrium is the special case where every σ_i^* puts probability 1 on a single action.

Support and best responses

Proposition

A profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a mixed Nash equilibrium *if and only if*, for every player $i \in N$, every pure action in the support of σ_i^* is a best response to σ_{-i}^*:

$$supp(\sigma_i^*) \subseteq BR_i(\sigma_{-i}^*) \qquad for\ every\ i \in N.$$

where $supp(\sigma_i) = \{ a_i \in A_i : \sigma_i(a_i) > 0 \}$ is the set of actions player i plays with positive probability.

Beyond finite games

This characterization is completely general: it holds in *any* game, including ones with infinite actions.

More drumroll, more Gregorian chants!

Existence of Nash equilibrium

Theorem (Nash, 1950)

Every *finite game* $\Gamma = (N, (A_i)_{i\in N}, (c_i)_{i\in N})$, that is one with finitely many players, each holding a finite action set A_i, has at least one mixed Nash equilibrium:

$$\exists\, \sigma^* = (\sigma_1^*, \ldots, \sigma_n^*) \in \Delta(A_1) \times \cdots \times \Delta(A_n)$$



John von Neumann

That's trivial



John Nash, Nobel Laureate (1994)

Image from: https://www.wikipedia.org/

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Existence in infinite games

Theorem (Glicksberg, 1952)

Let $\Gamma = (N, (A_i)_{i \in N}, (c_i)_{i \in N})$ be a game with N finite, where every A_i is a non-empty *compact* subset of a metric space, and every cost function $c_i : A \to \mathbb{R}$ is *continuous*. Then Γ has at least one mixed Nash equilibrium.

$$\exists\, \sigma^* = (\sigma_1^*, \ldots, \sigma_n^*) \in \Delta(A_1) \times \cdots \times \Delta(A_n)$$

each σ_i^* is now a probability measure on A_i, rather than a finite list of probabilities.

Rock, Paper, Scissors

- We can assume every player uses *all three actions,* otherwise the opponent could win for certain, by never playing the one action that beats it.
- Every action played is then in the support, so by the *Proposition*, R, P and S must all be best responses. This means player 1's expected cost is the same for each:

$$c_1(R, \sigma_2) = c_1(P, \sigma_2) = c_1(S, \sigma_2)$$

- Writing $\sigma_2 = (q_{R,2}, q_{P,2}, q_{S,2})$ for player 2's mixing probabilities, this indifference condition becomes:

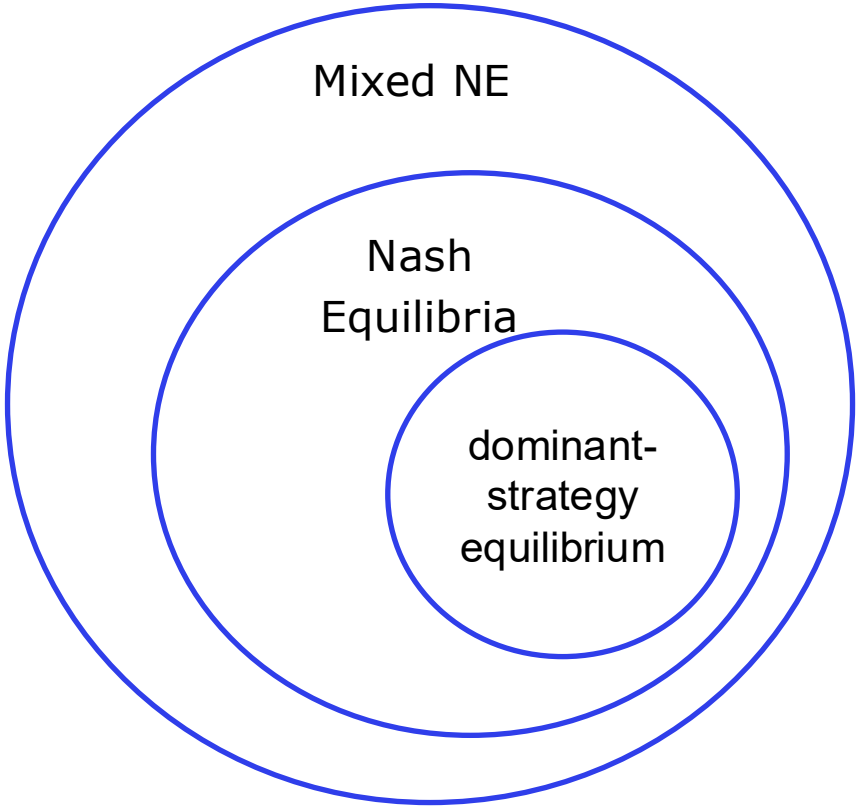
$$-q_{P,2} + q_{S,2} = q_{R,2} - q_{S,2}$$
$$q_{R,2} - q_{S,2} = -q_{R,2} + q_{P,2}$$
$$q_{R,2} + q_{P,2} + q_{S,2} = 1$$

- Solving this system gives $q_{R,2} = q_{P,2} = q_{S,2} = 1/3$. The same argument for player 1 shows the unique mixed Nash equilibrium has both players randomizing uniformly over Rock, Paper and Scissors.

	R	P	S
R	0, 0	-1, 1	1, -1
P	1, -1	0, 0	-1, 1
S	-1, 1	1, -1	0, 0

Mixed Nash equilibrium, fact sheet

- If a game fulfills the condition of either one of the theorems, it has at least one mixed Nash equilibrium
- To find it is PPAD hard (not in polynomial time)
- To check if there are two is NP-complete
- The set of mixed Nash equilibria is not arbitrary, but ugly




DTU